Supervised Learning

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Refreshing Mathematical Tools

Learning goals

- Refresher on the basics of probability theory

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PROBABILITY SPACE / 2

Examples:

- *Coin Tossing* — Possible outcomes are $\Omega = \{\mathtt{H}, \mathtt{T}\}$ with $\mathtt{H}$ resp. $\mathtt{T}$ representing "heads" resp. "tails". The allowable events are contained in $\mathcal{F} = \{\varnothing, \{\mathtt{H}\}, \{\mathtt{T}\}, \{\mathtt{H}, \mathtt{T}\}\}$. If the coin is fair, then $\mathbb{P}(\mathtt{H}) = \mathbb{P}(\mathtt{T}) = 1/2$.
- *Dice Rolling*— Possible outcomes are $\Omega = \{1, 2, 3, 4, 5, 6\}$. The allowable events are contained in 2^{Ω}, i.e., the power set of Ω. If the dice is fair, then $\mathbb{P}(i) = 1/6$ for $i = 1, \ldots, 6$.

Further properties. For any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ the following properties hold

- *Monotonicity* — $A, B \in \mathcal{F}$, and $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- *Union bound* — For any finite or countably infinite sequence events $E_1, E_2, \ldots,$

$$\mathbb{P}(\cup_{i \geq 1} E_i) \leq \sum_{i \geq 1} \mathbb{P}(E_i).$$

INDEPENDENCE / 2

Conditional probability. The *conditional probability* that event $A \in \mathcal{F}$ occurs given that event $B \in \mathcal{F}$ occurs is $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. The conditional probability is well-defined only if $\mathbb{P}(B) > 0$.

Bayes rule. For two events $A, B \in \mathcal{F}$ it holds that

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

The law of total probability. Let $E_1, \ldots, E_n \in \mathcal{F}$ be mutually disjoint events, such that $\cup_{i=1}^{n} E_i = \Omega$, then $\forall A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap E_i) = \sum_{i=1}^{n} \mathbb{P}(A|E_i)\mathbb{P}(E_i).$$

RANDOM VARIABLES / 2

- Functions of random variables are again random variables, i.e., if $f : \mathbb{R} \to \mathbb{R}$ is some (measurable) function, then $Z = f(X)$ is also a random variable.
- Identically distributed — Two random variables X and Y are identically distributed if their probability distributions coincide, i.e., $\mathbb{P}_X = \mathbb{P}_Y$.

One distinguishes between two types of random variables:

- A *discrete random variable* is a random variable that can take only a finite or countably infinite number of values. Its probability distribution is determined by the *probability mass function* which assigns a probability to each value in the image of X.
- A *continuous random variable* is a random variable which can take uncountably infinite number of values. Usually its probability distribution is determined by a *density function*, which assigns probabilities to intervals of the image of X.

DISCRETE RANDOM VARIABLES

If the image Ω_X of X is discrete (e.g., finite or countably infinite), then X is called a *discrete RV*.

For a discrete RV X, the function

$$p: \Omega_X \to [0,1],\ x \mapsto \mathbb{P}(X \in \{x\}) = \mathbb{P}(X = x)$$

is called a probability function or probability mass function of X.

Obviously, $p(x) \geq 0$ and $\sum_{x \in \Omega_X} p(x) = 1$.
Examples:

- *Bernoulli distribution*: For a binary RV with $\Omega_X = \{0, 1\}$, $X \sim \mathrm{Ber}(\theta)$ if $p(1) = \theta$ and $p(0) = 1 - \theta$.
- *Binomial distribution*: $X \sim \mathrm{Bin}(n, \theta)$ if

$$p(k) = \begin{cases} \binom{n}{k}\theta^k(1-\theta)^{n-k} & \text{if } k \in \{0, \ldots, n\} \\ 0 & \text{otherwise} \end{cases}.$$

EXPECTED VALUE/EXPECTATION

Expectation is the most basic characteristic of a random variable. Let X be a random variable, then the expectation of X, denoted by $\mathbb{E}(X)$, is

$$\mathbb{E}(X) = \int x \, dF(x) = \begin{cases} \sum_{x \in \Omega_X} x \, p(x) & \text{if } X \text{ is discrete} \\ \int_{\Omega_X} x \, p(x) \, dx & \text{if } X \text{ is continuous} \end{cases}$$

provided the sum resp. the integral is well-defined and exists.

Some important properties of the expected value are:

- Linearity — For any constants $c_1, c_2 \in \mathbb{R}$ and any pair of random variables X and Y it holds that $\mathbb{E}(c_1 X + c_2 Y) = c_1 \mathbb{E}(X) + c_2 \mathbb{E}(Y)$.
- Transformations — If $f : \mathbb{R} \to \mathbb{R}$ is a (measurable) function, then the expectation of $f(X)$ is

$$\mathbb{E}(f(X)) = \int f(x) \, dF(x) = \begin{cases} \sum_{x \in \Omega_X} f(x) \, p(x) & \text{if } X \text{ is discrete} \\ \int_{\Omega_X} f(x) \, p(x) \, dx & \text{if } X \text{ is continuous} \end{cases}$$

(provided the sum resp. integral exists.)

VARIANCE AND COVARIANCE

The variance of a RV X is defined as follows:

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}(X))^2\right] = \int_{\Omega_X} (x - \mathbb{E}(X))^2 \, dF(x)\,,$$

provided the integral on the right-hand side exists.

The *standard deviation* is defined by $\sqrt{\mathrm{Var}(X)}$.

The *covariance* between RVs X and Y is

$$\mathrm{Cov}(X, Y) = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right]$$

MULTIVARIATE RANDOM VARIABLES

RVs $X_1, \ldots, X_n$ over the same probability space can be combined into a *random vector* $\boldsymbol{X} = (X_1, \ldots, X_n)$.

Their joint distribution is specified by the joint mass/density function $p_{\boldsymbol{X}}$, such that for any measurable set A it holds that

$$\mathbb{P}(\boldsymbol{X} \in A) = \begin{cases} \sum_{(x_1,\ldots,x_n)\in A} p_{\boldsymbol{X}}(x_1, \ldots, x_n) & \text{if } X_1, \ldots, X_n \text{ are discrete} \\ \int_A p_{\boldsymbol{X}}(x_1, \ldots, x_n)\, dx_1 \ldots dx_n & \text{if } X_1, \ldots, X_n \text{ are continuous} \end{cases}$$

The marginal distribution p_1 of X_1 is given by

$$p_1(x_1) = \begin{cases} \sum_{(x_2,\ldots,x_n)\in\Omega_{X_2}\times\ldots\times\Omega_{X_n}} p_{\boldsymbol{X}}(x_1, \ldots, x_n) & \text{if discrete} \\ \int_{\Omega_{X_2}\times\ldots\times\Omega_{X_n}} p_{\boldsymbol{X}}(x_1, \ldots, x_n)\, dx_2 \ldots dx_n & \text{if continuous} \end{cases}$$

In the same way, the marginal distributions of $X_2, \ldots, X_n$ are defined. The same type of projection (summation/integration over all remaining variables) is used to define marginal distributions on subsets of variables $(X_{i_1}, \ldots, X_{i_k})$ with $\{i_1, \ldots, i_k\} \subseteq \{1, \ldots, n\}$.

INDEPENDENCE OF RANDOM VARIABLES / 2

Some important properties and concepts with respect to independence are:

- The iid assumption — Random variables $X_1, \ldots, X_n$ are called *independent and identically distributed* (iid) iff they are mutually independent and each random variable has the same probability distribution as the others.
- Independence under transformations — Let X and Y be independent random variables and $f, g : \mathbb{R} \to \mathbb{R}$ are (measurable) functions. Then, $f(X)$ and $g(Y)$ are independent as well.

CONDITIONAL DISTRIBUTIONS

If (X, Y) have a joint distribution with mass function $p_{X,Y}$, then the *conditional probability mass function* for X given Y is defined by

$$p_{X|Y}(x \mid y) = \frac{p(X = x, Y = y)}{p(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

provided $p(Y = y) > 0$.

Likewise, in the continuous case, the *conditional probability density function* is given by

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

provided $p_Y(y) > 0$. Then,

$$p(X \in A \mid Y = y) = \int_A p_{X|Y}(x \mid y)\, dx\,.$$

The soundness of this definition is less obvious than in the discrete case, due to conditioning on an event of probability 0 here.

CONDITIONAL EXPECTED VALUE/EXPECTATION / 2

Some important properties of the conditional expected value are the following. For any random variables X, Y, Z it holds that

- Linearity — For any constants $c_1, c_2 \in \mathbb{R}$ it holds that $\mathbb{E}(c_1 X + c_2 Y|Z) = c_1 \mathbb{E}(X|Z) + c_2 \mathbb{E}(Y|Z)$.
- Independence — If X and Y are independent random variables, then $\mathbb{E}(X|Y) = \mathbb{E}(X)$.
- Transformations — If $f : \mathbb{R} \to \mathbb{R}$ is a (measurable) function, then the conditional expectation of $f(X)$ given $Y = y$ is

$$\mathbb{E}(f(X)|Y = y) = \begin{cases} \sum_{x \in \Omega_X} f(x)\, p_{X|Y}(x \mid y) & \text{discrete case} \\ \int_{\Omega_X} f(x)\, p_{X|Y}(x \mid y)\, dx & \text{continuous case} \end{cases}$$

- Law of total expectation — $\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$.
- Tower property — $\mathbb{E}(\mathbb{E}(X|Y, Z)|Y) = \mathbb{E}(X|Y)$.